

IT Project Management AI-Enhanced Delivery and Control

BSIT LECTURE HANDOUT

Course: Data Science Program: BSIT

Planning • Estimation • Risk • Agile Delivery • AI Decision Support

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1 Introduction to IT Project Management

IT Project Management

IT project management is the structured planning, execution, monitoring, and closure of technology-related work so that a defined product, service, or system is delivered within agreed scope, time, cost, quality, and risk constraints.

Why This Matters in a Data Science Course

Data science projects are still projects. They require scope definition, stakeholder communication, resource planning, schedules, budgets, quality control, and risk management. What changes in this handout is the use of artificial intelligence to improve how managers forecast work, detect risk, summarize progress, and support decisions.

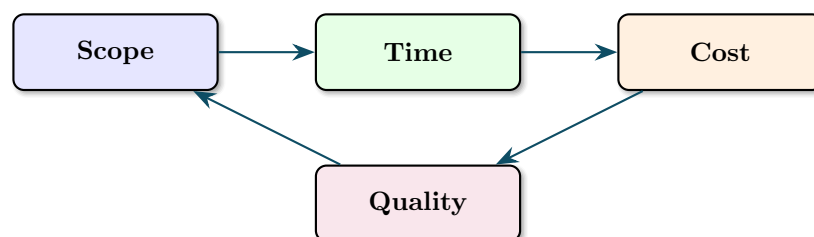
1.1 Project, Program, and Operations

- **Project:** Temporary effort with a start and end date that produces a unique output.
- **Program:** Group of related projects managed together for coordinated benefit.
- **Operations:** Ongoing repetitive work that keeps an organization running.

1.2 Typical IT Project Goals

- Deliver a software application or platform.
- Implement infrastructure, cybersecurity, or data solutions.
- Improve service quality, automation, or reporting.
- Meet regulatory, business, or academic requirements.

1.3 The Triple Constraint and Beyond

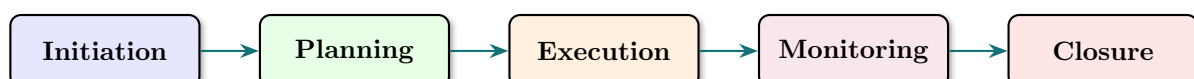


Management Reality

Modern project management goes beyond the classic scope-time-cost triangle. It also considers quality, stakeholder satisfaction, business value, security, sustainability, and team capacity. AI can help observe these dimensions, but it does not remove the need for managerial judgment.

2 The Project Lifecycle and Management Foundations

2.1 Project Lifecycle



2.2 Core Management Activities by Phase

Phase	Typical Activities
Initiation	Identify problem, stakeholders, goals, feasibility, constraints, and success criteria
Planning	Define scope, work breakdown, schedule, budget, risks, resources, communication, and quality approach
Execution	Build deliverables, coordinate team members, manage vendors, resolve blockers, and track work
Monitoring and control	Measure progress, compare against baseline, manage changes, update risks, and report status
Closure	Obtain acceptance, document lessons learned, archive records, and transition outputs to operations

2.3 Key Knowledge Areas

- Scope management
- Schedule management
- Cost management
- Quality management
- Risk management
- Resource management
- Communication management
- Stakeholder management

Important Principle

If a project team adopts AI tools but ignores project fundamentals such as scope control, stakeholder alignment, and risk review, the project can still fail. AI strengthens management practice; it does not replace it.

3 What AI Changes in Project Management

3.1 AI-Enhanced Project Management

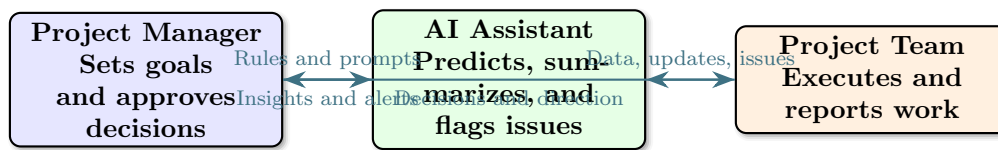
AI-Based Project Management

AI-based project management refers to the use of machine learning, natural language processing, predictive analytics, automation, and intelligent assistants to support planning, monitoring, decision-making, and communication throughout the project lifecycle.

3.2 Where AI Adds Value

- Forecasting schedule delays from historical patterns.
- Estimating effort, duration, or budget using past project data.
- Detecting risk signals from issue trackers, email summaries, or sprint histories.
- Automatically summarizing meetings, action items, and progress updates.
- Classifying change requests and support tickets.
- Recommending task prioritization or resource allocation.

3.3 Human Manager Plus AI Assistant



3.4 Benefits and Limits

Potential Benefits	Practical Limits
Faster reporting and summarization	Poor results when input data is incomplete or noisy
Earlier detection of schedule and cost risk	Models may inherit bias from past project history
Better visibility into team workload	AI can recommend actions without understanding organizational politics
Improved consistency in documentation	Sensitive project data may raise privacy or compliance concerns
Automation of repetitive coordination tasks	Final accountability still belongs to human managers

4 AI Across Planning, Execution, and Control

4.1 AI in Planning

- Generate draft work breakdown structures from project objectives.
- Suggest milestones, dependencies, and probable delivery dates.
- Estimate effort by comparing the current project to similar historical projects.
- Identify missing requirements or ambiguous scope statements from project documents.

4.2 AI in Execution

- Summarize stand-up meetings and extract action items.
- Recommend task assignments based on skill patterns and availability.
- Prioritize backlog items using urgency, impact, and dependency signals.
- Monitor project communication streams for blockers, escalation signals, or unresolved issues.

4.3 AI in Monitoring and Control

- Predict whether milestones will be missed.
- Detect cost variance trends earlier than manual review.
- Analyze defect patterns and quality risk indicators.
- Produce dashboard narratives in plain language for managers and stakeholders.

Simple AI PM Workflow

A team uses issue tracker data, sprint velocity, meeting transcripts, and timesheet summaries as inputs. An AI service generates a weekly report that highlights late tasks, probable schedule slippage, unresolved risks, and missing approvals. The project manager reviews the output before sharing it with stakeholders.

5 Agile, Scrum, and Hybrid AI-Supported Delivery

5.1 Why Agile Fits Many IT Projects

Adaptive Delivery

Many IT projects, especially software and analytics projects, face changing requirements. Agile methods address this by delivering incrementally, gathering feedback frequently, and re-prioritizing based on value and learning.

5.2 Common Agile Roles and Artifacts

Element	Purpose	AI Support Example
Product backlog	Ordered list of desired work	Suggest item grouping, deduplicate similar stories
Sprint backlog	Work selected for current sprint	Flag overload based on historical velocity
Scrum Master	Removes blockers and supports process	Summarize impediments across stand-ups
Product Owner	Prioritizes value and requirements	Analyze feedback themes from users or stakeholders
Burndown chart	Visualizes remaining work	Add natural-language explanation of trend anomalies

5.3 Hybrid Project Management

- Some parts of a project may need predictive planning, such as procurement, security review, or compliance milestones.
- Other parts may need Agile iteration, such as interface design, analytics features, or integration refinement.
- AI can support both by improving forecasts, summarizing progress, and highlighting deviations.

Managerial Caution

AI recommendations should not be treated as unquestionable truth. If the backlog is poorly written, if stakeholders change their minds, or if the team composition shifts, an AI-generated sprint plan can become misleading very quickly.

6 Data, Metrics, and Dashboards for AI Project Management

6.1 What Data AI Needs

- Historical schedules and milestone completion data.

- Task durations, effort logs, and velocity records.
- Risk registers, issue logs, and change requests.
- Meeting notes, chat summaries, and decision records.
- Quality indicators such as defects, rework, and test outcomes.

6.2 Important Performance Metrics

Metric	What It Shows	AI Use
Schedule variance	Difference between planned and actual timeline	Predict future delay risk
Cost variance	Difference between budgeted and actual cost	Flag overspending patterns
Velocity	Team delivery rate in Agile work	Forecast sprint capacity
Defect density	Quality problems relative to output size	Identify unstable modules or teams
Lead time	Time from request to completion	Detect bottlenecks in workflow

6.3 Dashboard Design Principles

- Show only the most decision-relevant indicators.
- Use trends and comparisons, not raw counts alone.
- Separate confirmed issues from AI-predicted risks.
- Keep audit trails so managers can explain why a recommendation was generated.

6.4 Interpretability Matters

💡 Explainable Insight

If an AI system predicts that a project will miss its deadline, the project manager should know why. Useful explanations might include reduced team velocity, late approvals, unresolved blockers, increasing defect count, or repeated scope changes.

7 Risk, Governance, and Ethics in AI-Based Project Management

7.1 Project Risks Introduced by AI

- Overreliance on AI-generated estimates.
- Leakage of confidential project data into external tools.
- Biased recommendations caused by poor historical data.
- Hallucinated summaries or inaccurate status interpretations.
- False confidence in automated prioritization.

7.2 Governance Controls

- Define which project data AI tools are allowed to access.
- Keep a human approval step for major decisions.
- Review prompt templates, model behavior, and security controls regularly.
- Log outputs that affect budget, schedule, scope, or staffing decisions.
- Train teams to challenge AI outputs instead of blindly accepting them.

7.3 Ethical Questions

- Is the AI making recommendations that unfairly favor or penalize certain team members?
- Is the system using personal productivity data in ways employees did not expect?
- Are stakeholders clearly informed when reports or summaries are AI-generated?
- Can a project team explain and justify a decision that was influenced by AI?

Principle of Accountability

AI may assist with analysis, but responsibility for project outcomes stays with human managers and sponsoring organizations. Accountability cannot be delegated to a model.

8 Case Study and BSIT Skills Roadmap

8.1 Mini Case Study: AI-Supported Student Information System Project

Scenario

A BSIT team is assigned to build a student information dashboard for a university. The team uses Agile delivery with two-week sprints. An AI assistant helps draft user stories, summarize sprint reviews, detect repeated blockers from stand-up notes, and forecast whether the final release date is at risk.

Project management lessons from the case:

- Human leaders still define scope, approve priorities, and resolve tradeoffs.
- AI speeds up reporting and surfacing risk, but team data quality determines usefulness.
- Sensitive student data requires strong governance over AI tools and logs.
- The most useful AI outputs are often summaries, risk signals, and planning suggestions rather than full automation.

8.2 Skills BSIT Students Should Build

- | | |
|------------------------------|---------------------------------------|
| • Project charter writing | • Dashboard interpretation |
| • Work breakdown structuring | • Prompting AI assistants responsibly |
| • Gantt and sprint planning | • Data privacy awareness |
| • Risk register design | • Team communication and reporting |

8.3 Professional Mindset

💡 AI as Decision Support

The best project managers do not ask AI to think for them. They use AI to surface patterns faster, reduce manual reporting effort, and improve visibility while keeping control over judgment, ethics, communication, and accountability.

9 Key Takeaways and Review Questions

9.1 Key Takeaways

- IT project management organizes technology work so it can be delivered with control over scope, time, cost, quality, and risk.
- AI can strengthen planning, forecasting, reporting, and risk detection across the project lifecycle.
- Good data, human oversight, and governance are necessary for AI-based project management to be trustworthy.
- Agile and hybrid delivery models can benefit from AI-supported summaries, predictions, and prioritization assistance.
- BSIT students should learn both classical project management foundations and responsible use of AI tools.

9.2 Review Questions

1. What is the difference between a project and ongoing operations?
2. How can AI improve planning and monitoring in IT projects?
3. Why should managers be cautious about AI-generated estimates and recommendations?
4. What kinds of project data are useful for AI-based forecasting?
5. How can Agile teams use AI without giving up human control?
6. Why is governance necessary when AI tools process project information?

🔗 Short Reflection Task

Imagine that your BSIT class is developing a campus mobile application. Describe one way AI could help during planning, one way it could help during execution, and one governance rule you would apply before letting the AI analyze project data.

End of Handout

IT Project Management — BSIT Program